

LECTURE 10

DJANGO DEMO

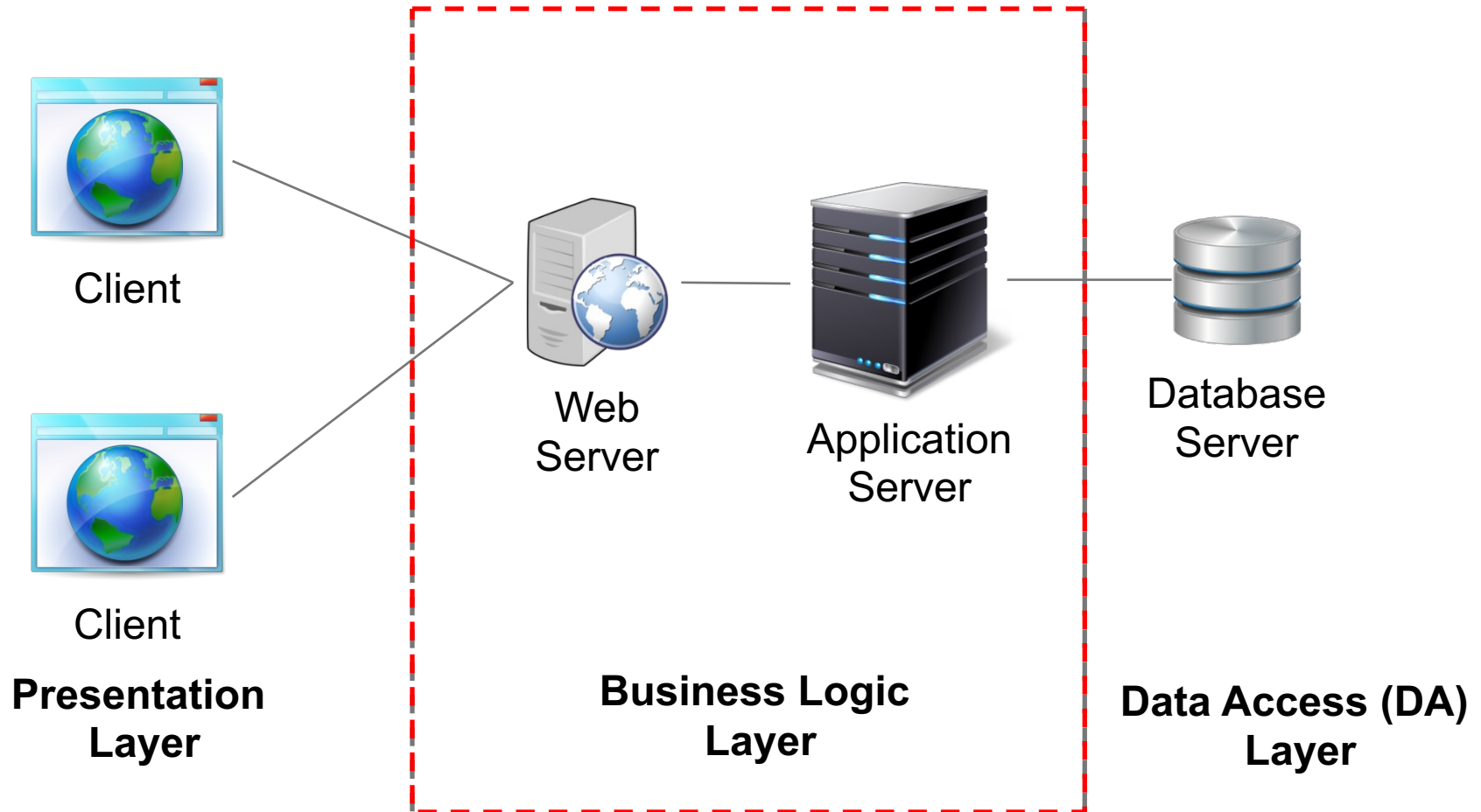
LEK HSIANG HUI

LEARNING OBJECTIVES

At the end of this lecture, you should understand:

- The basic design of enterprise systems architecture
- Appreciation of Django

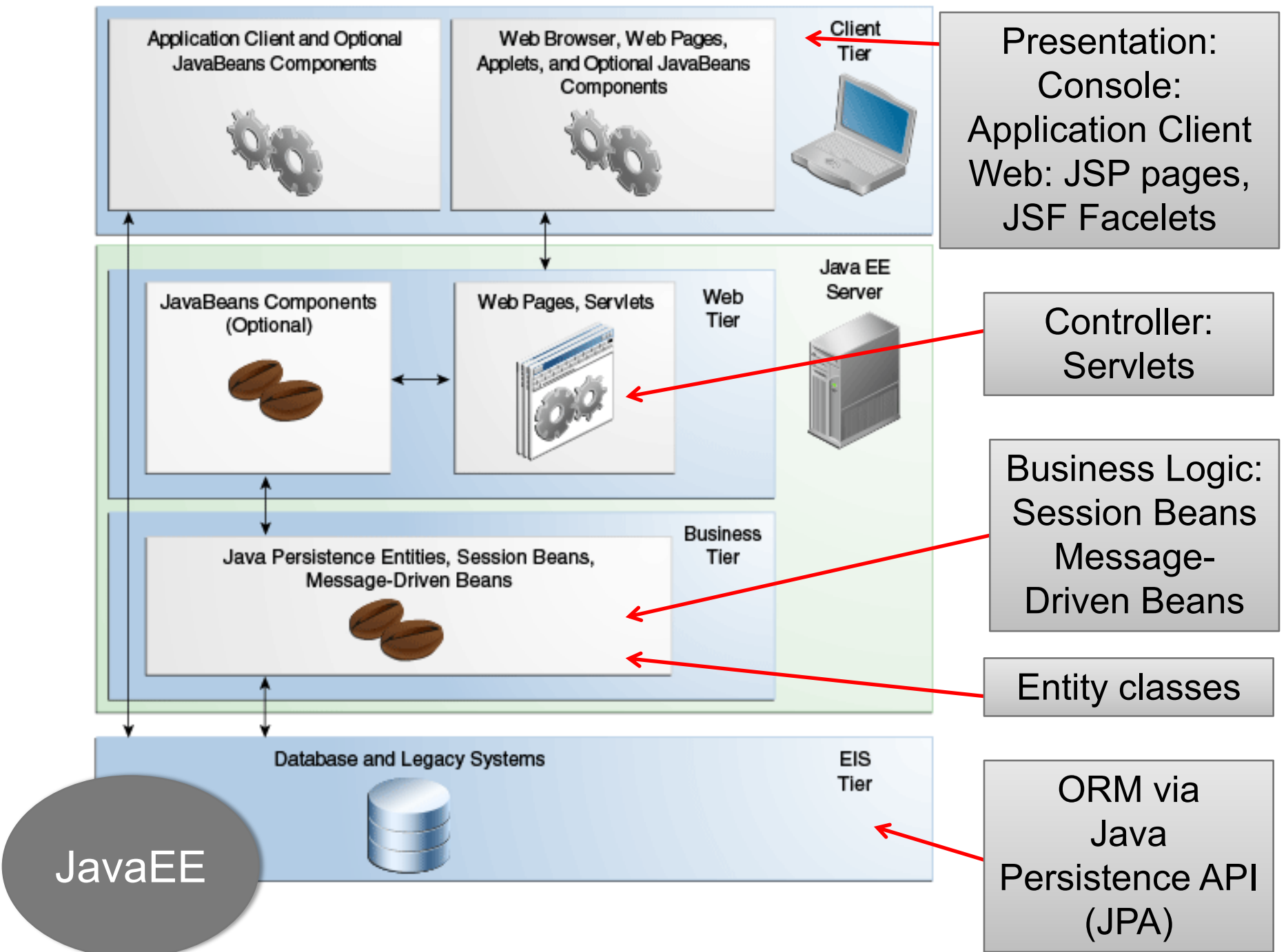
RECAP: MULTI-TIER ARCHITECTURE



ENTERPRISE SYSTEMS DEVELOPMENT (JAVA)

Enterprise systems development often involves using an enterprise development framework

- E.g. Java Enterprise Edition (Java EE) is a platform for building enterprise systems for Java developers
- Use in conjunction with Java Standard Edition (Java SE)



ENTERPRISE SYSTEM DEVELOPMENT (PYTHON)

Traditional approach:

- Backend (e.g. Django framework)
- Frontend (e.g. Django templating system)

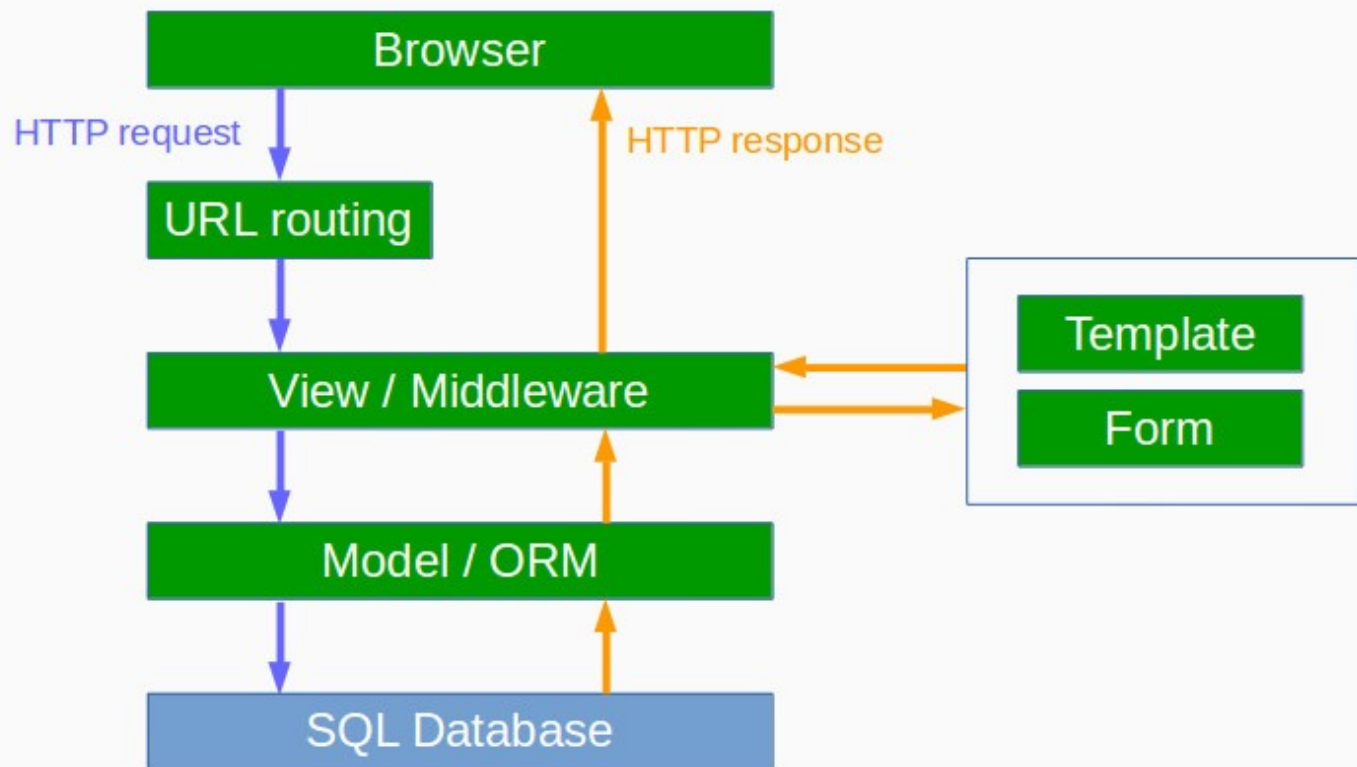
**Backend provides the core for frontend clients
(usually web) to access**

**Frontend clients provides the UI for accessing the
system**

Will focus mainly on this approach

DJANGO ARCHITECTURE

Django's architecture

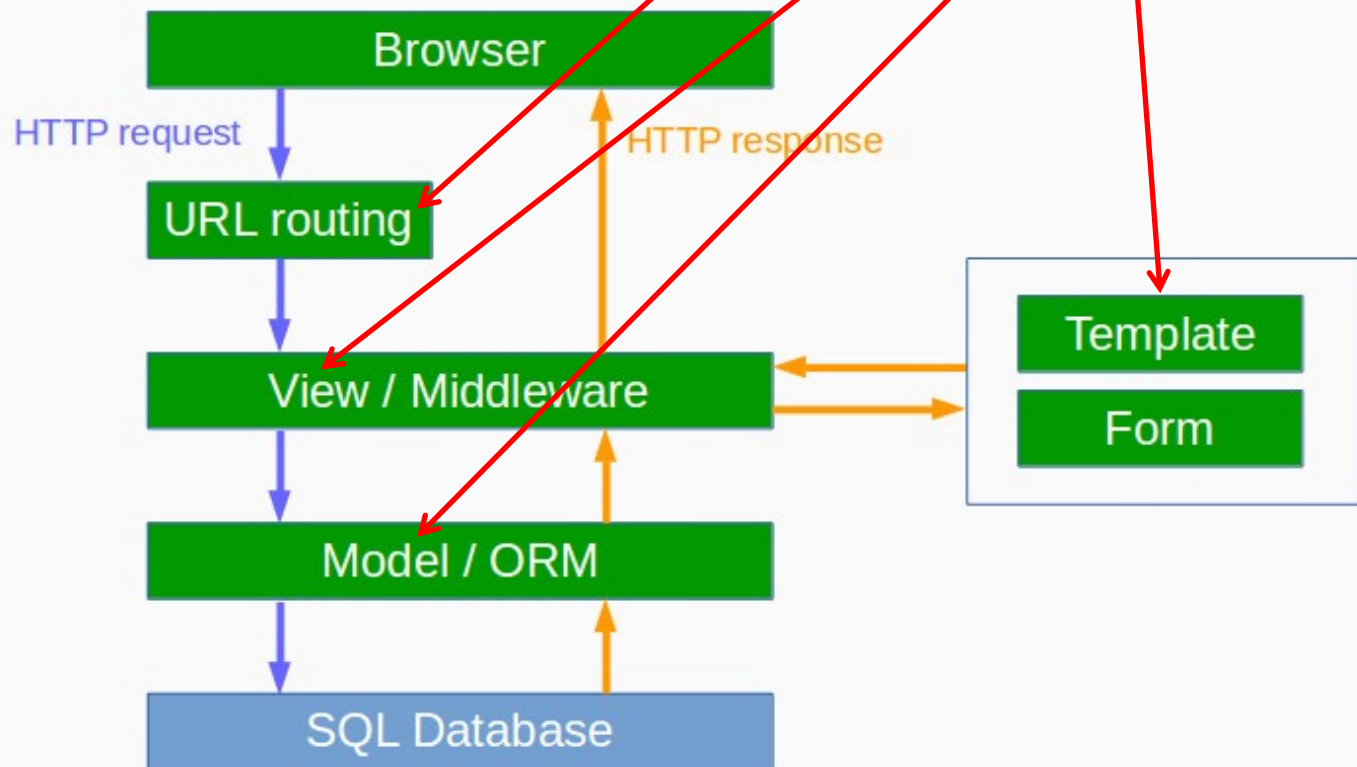


source: <https://fleschenberg.net/django-architecture-diagram/>

DJANGO ARCHITECTURE

Django's architecture

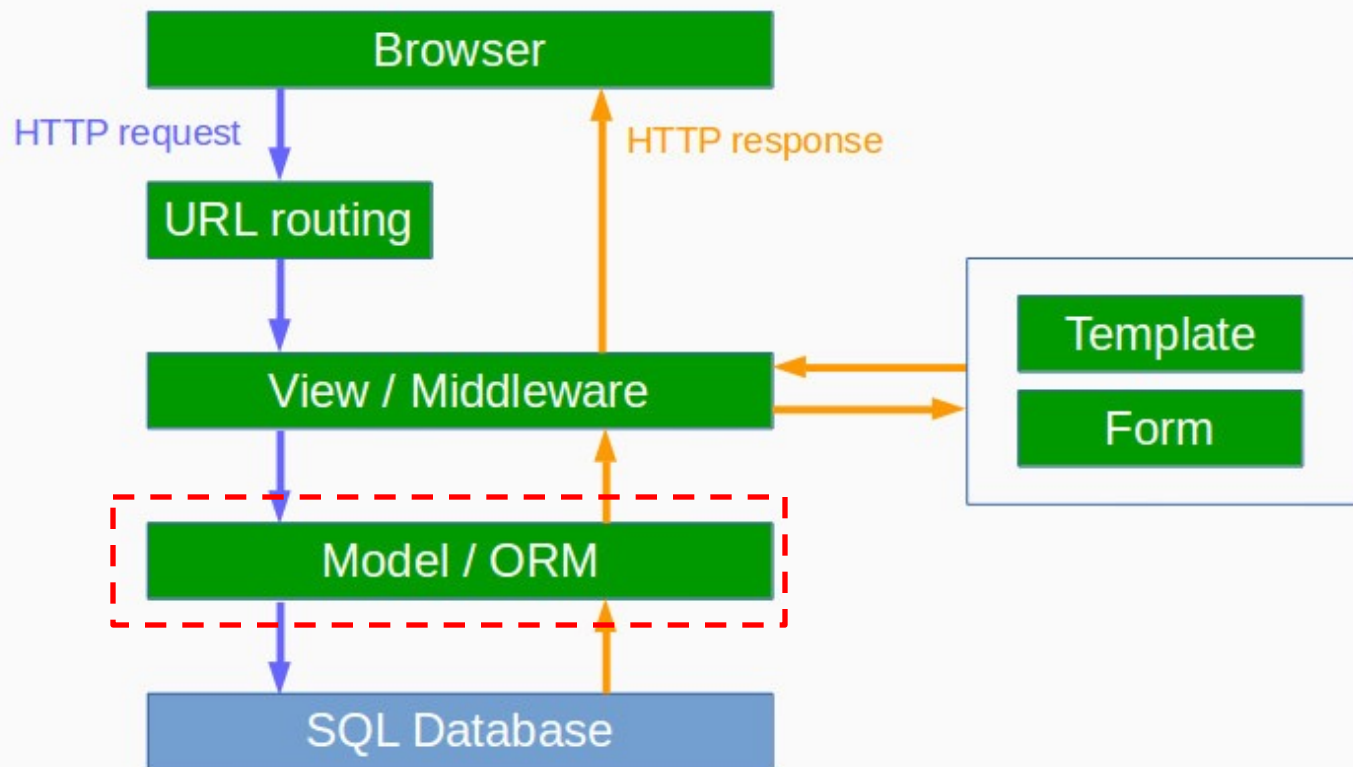
Separation of Concerns:
Model, View, Template,
URL routing (i.e.
Controller)



source: <https://fleschenberg.net/django-architecture-diagram/>

DJANGO ARCHITECTURE

Django's architecture



source: <https://fleschenberg.net/django-architecture-diagram/>

Inherits
`django.db.models.Model`

DEFINING MODEL

```
#models.py
```

```
from django.db import models
```

```
class Customer(models.Model):
```

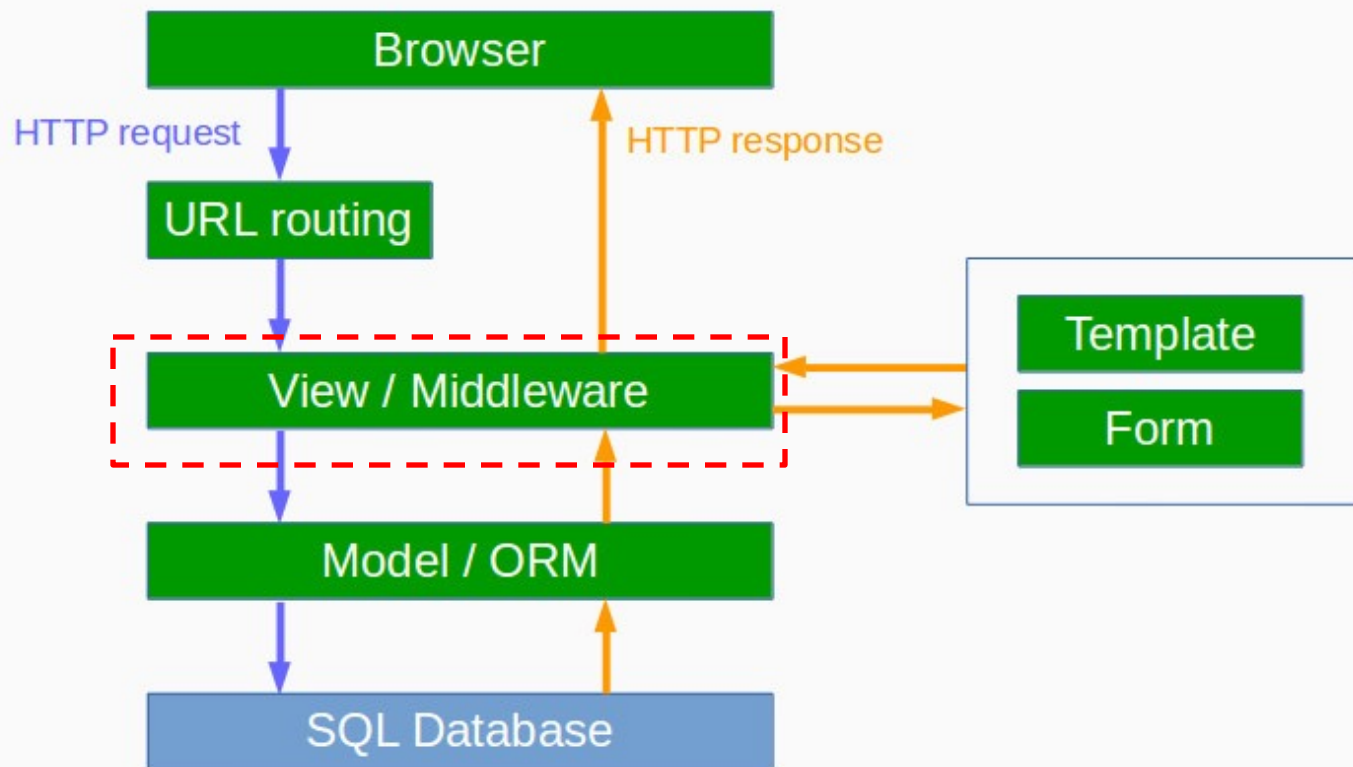
```
    name = models.CharField(max_length=30)
```

```
    age = models.IntegerField()
```

Add in the field names and field
type

DJANGO ARCHITECTURE

Django's architecture



source: <https://fleschenberg.net/django-architecture-diagram/>

Just normal Python
methods

DEFINING VIEW

`#views.py`

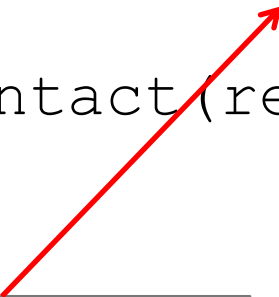
```
def about(request):  
    return render(request, 'aboutme.html',  
{'name': 'John Doe', 'age': 20})
```



```
def newContact(request):  
    ...
```

...

Data to pass to
Template



Template (page) to
display

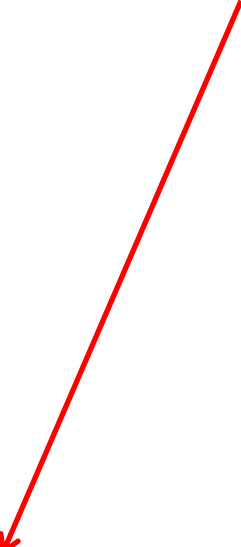


DEFINING VIEW

`#views.py`

```
...  
  
def list(request):  
    customers = Customer.objects.all()  
    return render(request, 'list.html',  
{'customers': customers})
```

Grab all the Customer objects (defined as Customer Model)



Data to pass to **list.html** Template

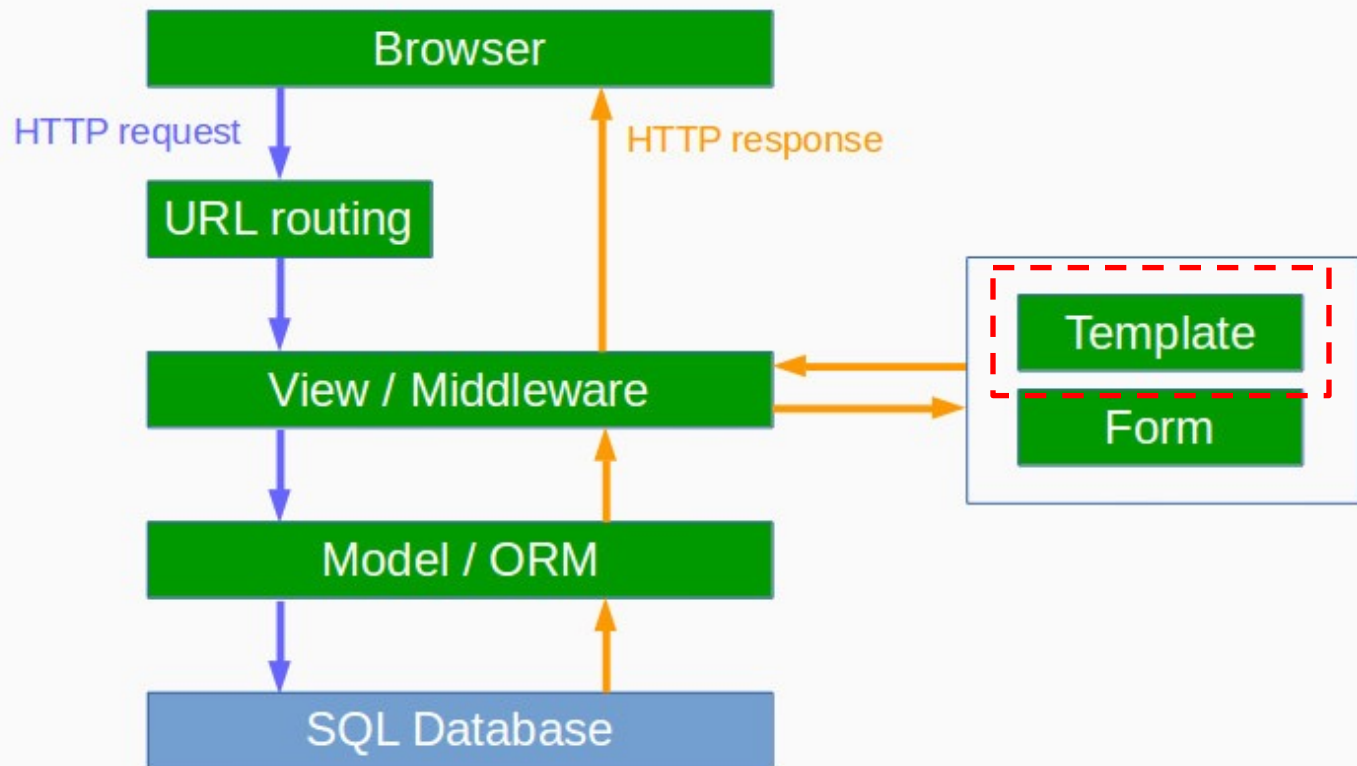


Template (page) to display



DJANGO ARCHITECTURE

Django's architecture



source: <https://fleschenberg.net/django-architecture-diagram/>

Just normal HTML
pages

DEFINING TEMPLATE

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>About me</title>
  </head>
  <body>
    <h1>Hi {{ name }}</h1>
    <h2>Age: {{ age }}</h2>
  </body>
</html>
```

▲ templates

- <> aboutme.html
- <> homepage.html
- <> list.html
- <> new.html

Display data passed
from the view function

DEFINING TEMPLATE

Just normal HTML
pages

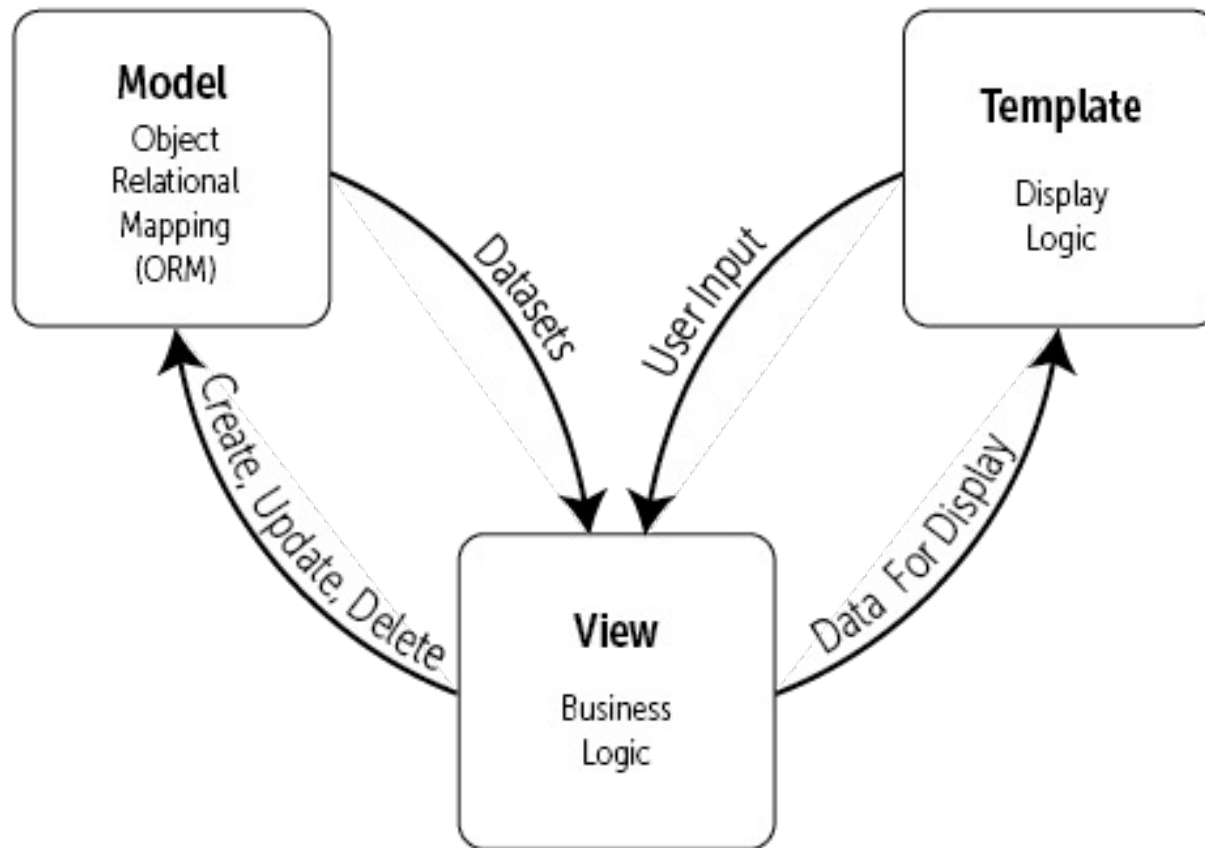
```
...
{% for customer in customers %}
<tr>
  <td> {{ customer.id }} </td>
  <td> {{ customer.name }} </td>
  <td> {{ customer.age }} </td>
  <td> <a
href="/delete?id={{customer.id}}">Delete</a> </td>
</tr>
{% endfor %}
...
```

templates

- <> aboutme.html
- <> homepage.html
- <> list.html
- <> new.html

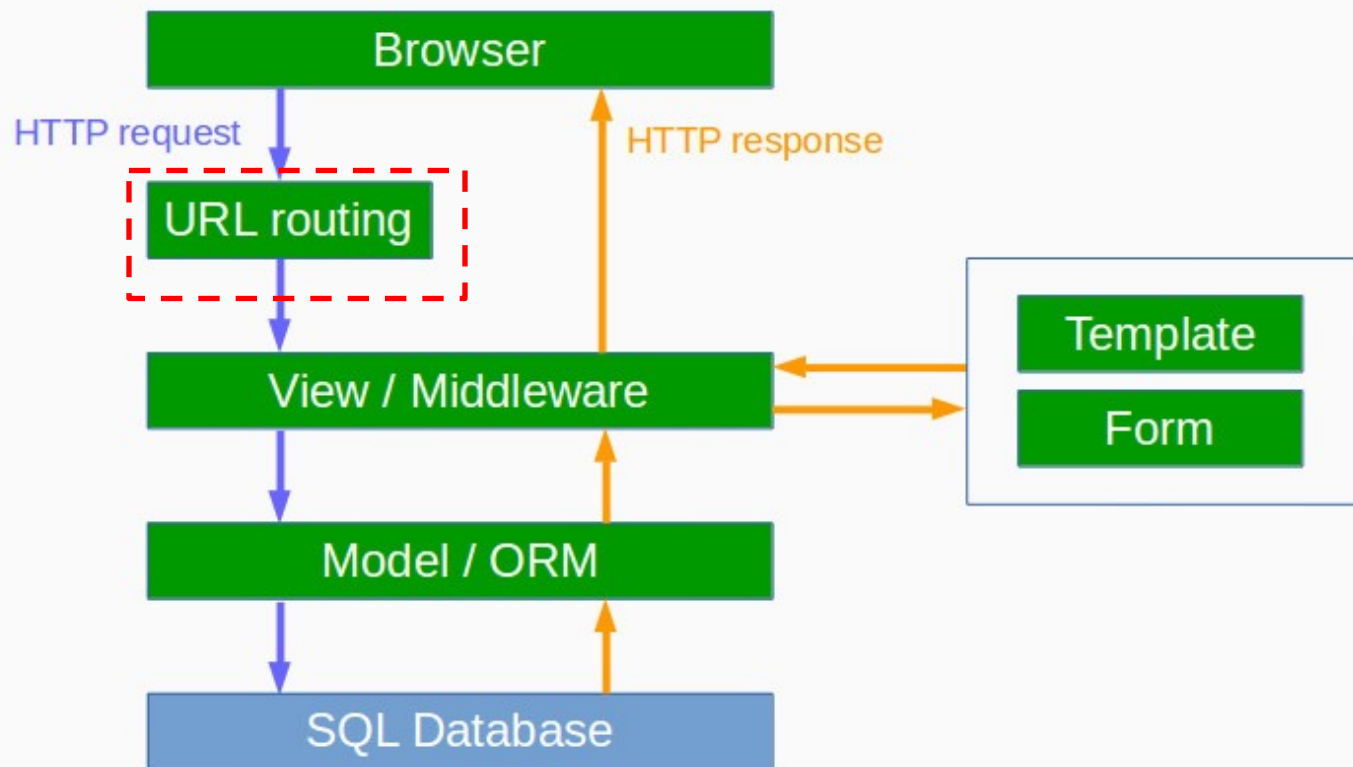
Iterate through the
data and display on
the page

MTV/MVT DESIGN PATTERN



DJANGO ARCHITECTURE

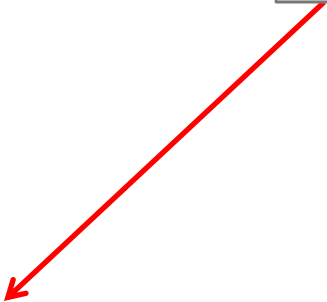
Django's architecture



source: <https://fleschenberg.net/django-architecture-diagram/>

DEFINING URL ROUTING (CONTROLLER)

Import
view.py



```
#urls.py
```

```
...
```

```
from crm.records import views as records_views
```

```
urlpatterns = [
```

```
    path('about/', records_views.about),
```

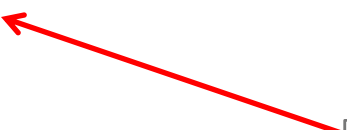
```
    path('new/', records_views.newContact),
```

```
    path('list/', records_views.list),
```

```
    path('delete/', records_views.deleteContact),
```

```
...
```

```
]
```



Linking each url endpoint to a
view function

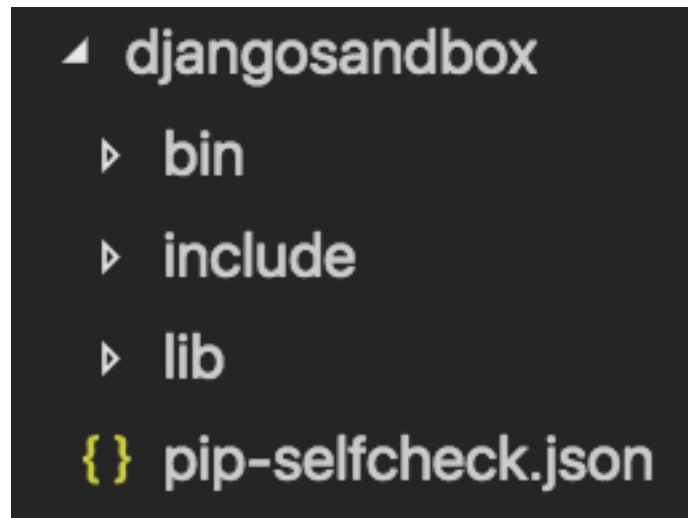
GETTING STARTED WITH DJANGO

Install virtualenv (optional)

- `pip install virtualenv`

Create virtual Python environment (django sandbox) with virtualenv (optional)

- `virtualenv --python=python3 django sandbox`



GETTING STARTED WITH DJANGO

Activate (djangosandbox) virtualenv

- `cd djangosandbox`
- `source bin/activate` (or just `Scripts\activate`)

```
NUSs-MacBook-Pro-93:test1 lekhsian$ cd djangosandbox/  
NUSs-MacBook-Pro-93:djangosandbox lekhsian$ source bin/activate
```



```
NUSs-MacBook-Pro-93:test1 lekhsian$ cd djangosandbox/  
NUSs-MacBook-Pro-93:djangosandbox lekhsian$ source bin/activate  
(djangosandbox) NUSs-MacBook-Pro-93:djangosandbox lekhsian$
```

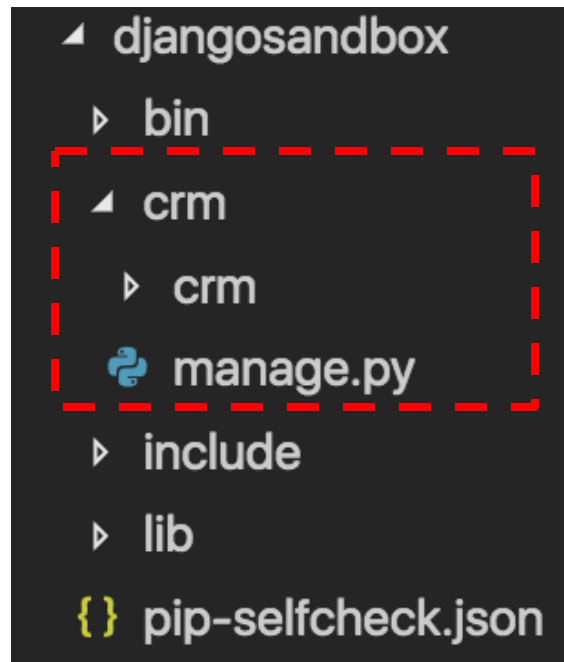
GETTING STARTED WITH DJANGO

Install Django (from the virtualenv)

- `pip install Django`

Start a Django project (crm)

- `django-admin startproject crm`



GETTING STARTED WITH DJANGO

Start Django server to test

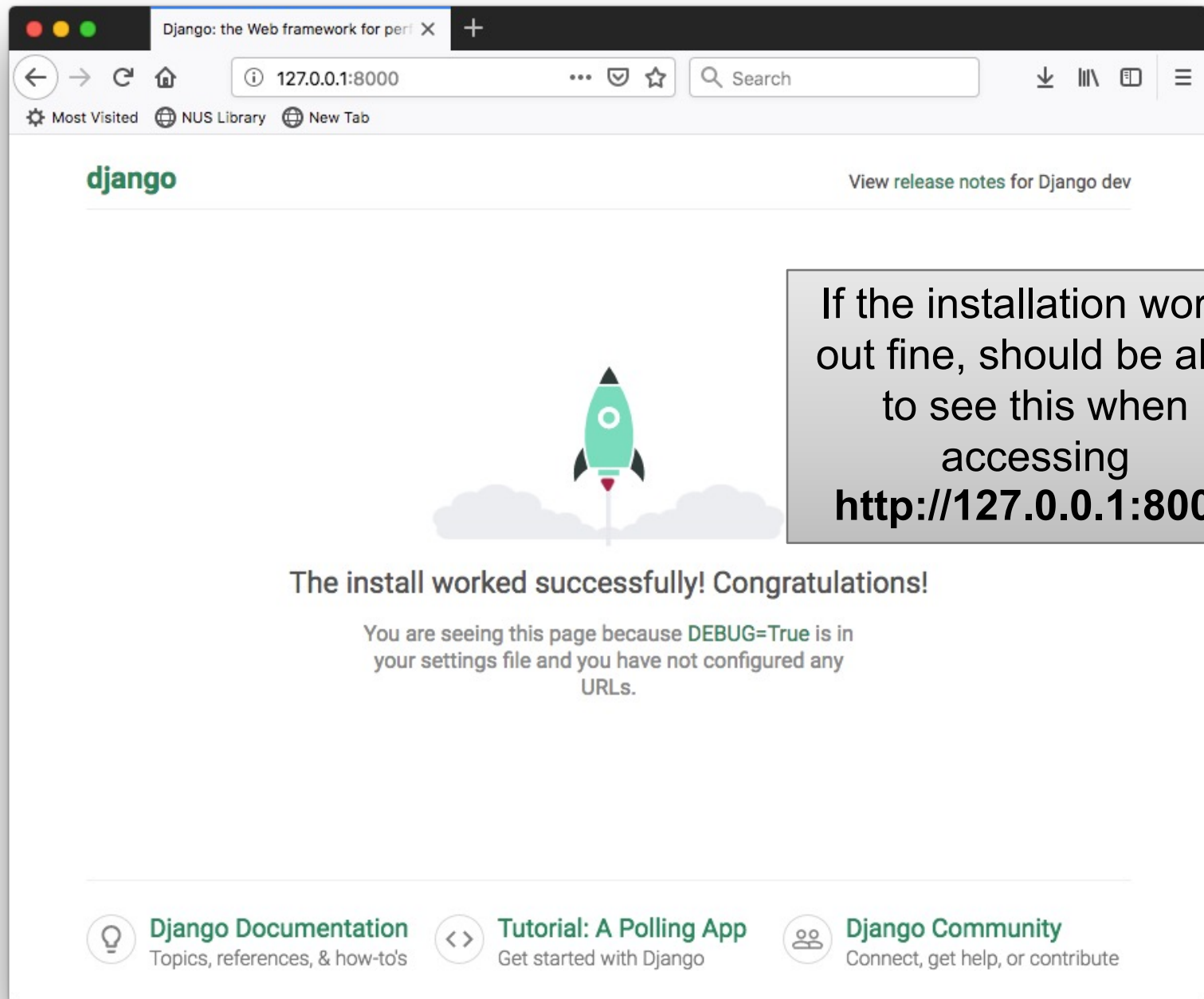
- `cd crm`
- `python manage.py runserver`

```
(djangosandbox) NUSs-MacBook-Pro-93:djangosandbox lekhsian$ cd crm/  
(djangosandbox) NUSs-MacBook-Pro-93:crm lekhsian$ python manage.py runserver  
Watching for file changes with StatReloader  
Performing system checks...
```

```
System check identified no issues (0 silenced).
```

```
You have 17 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.  
Run 'python manage.py migrate' to apply them.
```

```
March 12, 2019 - 08:51:28  
Django version 3.0, using settings 'crm.settings'  
Starting development server at http://127.0.0.1:8000/  
Quit the server with CONTROL-C.
```



GETTING STARTED WITH DJANGO

Run Django migrate operation to create the base database tables (using SQLite by default)

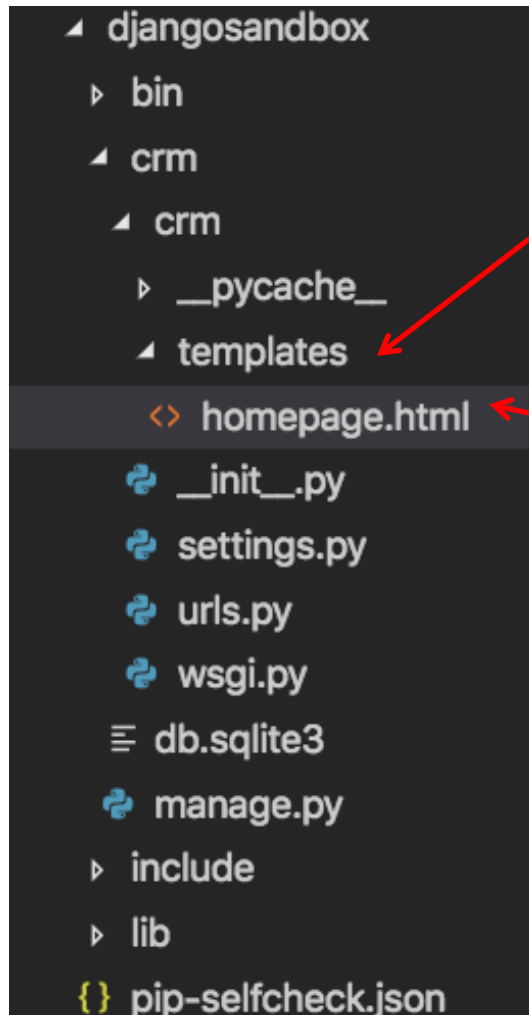
- `python manage.py migrate`

```
^C(djangosandbox) NUSs-MacBook-Pro-93:crm lekhsian$ python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying sessions.0001_initial... OK
(djangosandbox) NUSs-MacBook-Pro-93:crm lekhsian$ python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
March 12, 2019 - 08:57:13
Django version 3.0, using settings 'crm.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.
```

No more issues

GETTING STARTED WITH DJANGO



Create a new **templates** folder under **crm/crm**

Create a new file:
homepage.html

in the newly created
templates folder

Tell Django to find its templates from the **crm/crm/templates** folder

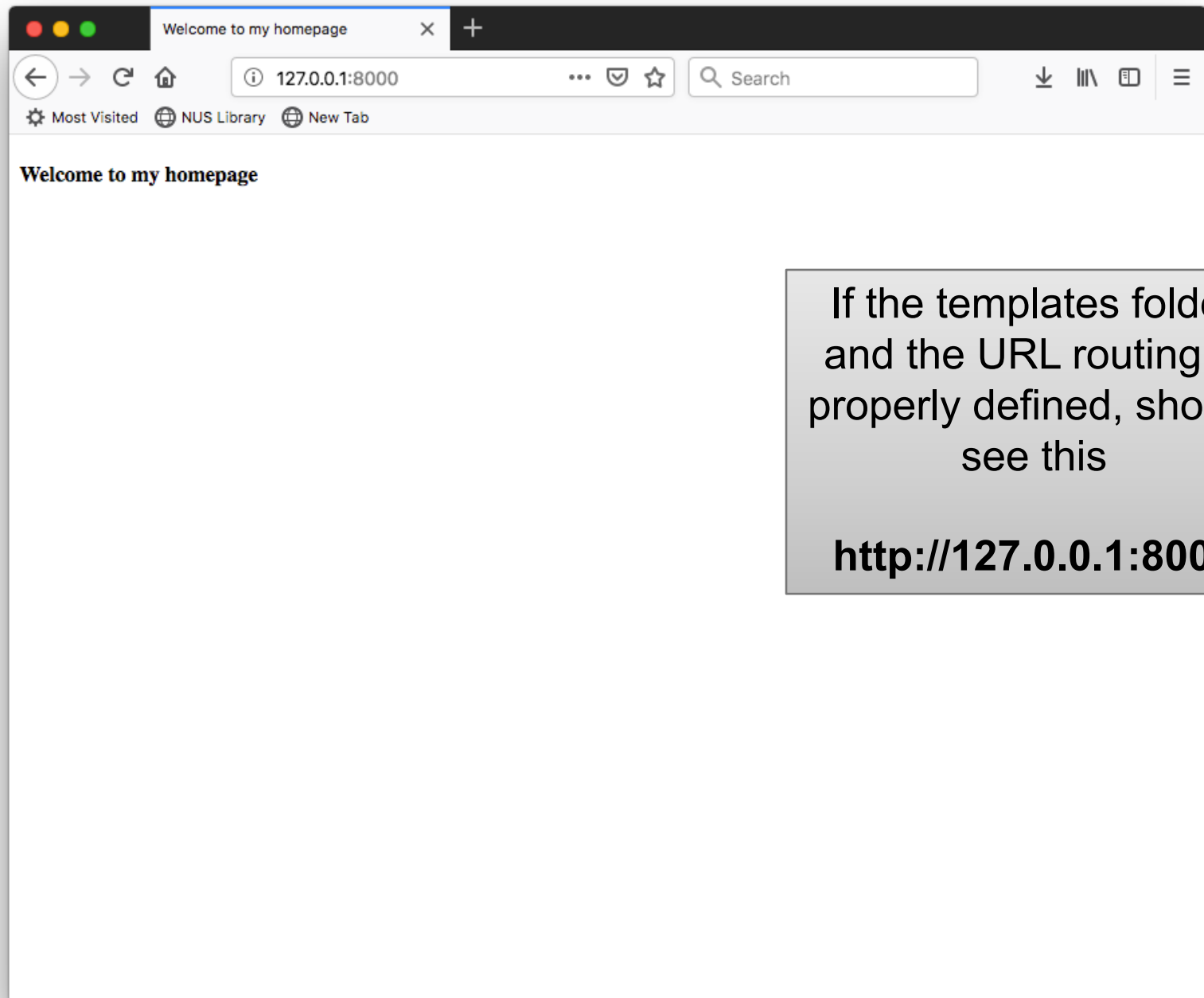


GETTING STARTED WITH DJANGO

```
...  
from django.contrib import admin  
from django.urls import path  
from django.views.generic import  
TemplateView  
  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('',  
    TemplateView.as_view(template_name='ho  
    mepage.html')),  
]
```

Map **http://127.0.0.1:8000/**
to **homepage.html**

```
└─ djangosandbox  
  └─ bin  
    └─ crm  
      └─ crm  
        └─ __pycache__  
          └─ templates  
            └─ <> homepage.html  
              └─ __init__.py  
                └─ settings.py  
                  └─ urls.py  
                    └─ wsgi.py  
                      └─ db.sqlite3  
                        └─ manage.py  
                          └─ include  
                            └─ lib  
                              └─ {} pip-selfcheck.json
```



If the templates folder
and the URL routing is
properly defined, should
see this

`http://127.0.0.1:8000`

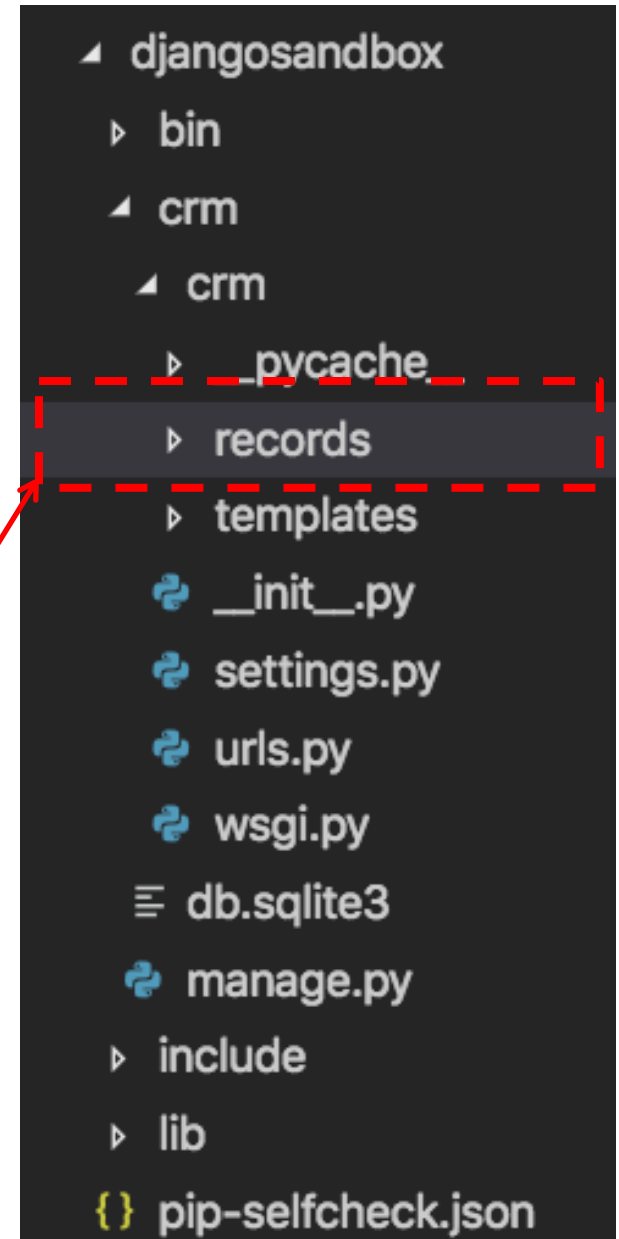
GETTING STARTED WITH DJANGO

Add a (records) Django app

- `django-admin startapp records`

A new **records** folder created with some generated files (e.g. **views.py**, **models.py**)

and make sure that the **records** folder is in the **crm/crm** folder (same level as **templates**)



GETTING STARTED WITH DJANGO

```
...  
...  
#Add the Django "records" app  
INSTALLED_APPS = [  
    ...  
    'crm.records',  
]  
...
```

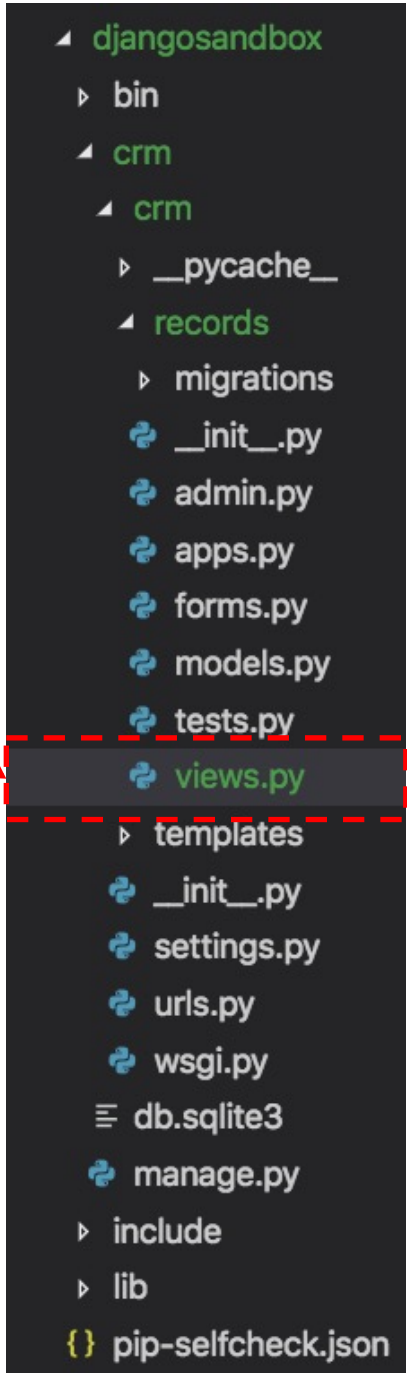
Tell Django about this new app so that **urls.py** will be able to import functions from the **views.py** in the records folder

- └─ djangosandbox
 - └─ bin
 - └─ crm
 - └─ crm
 - └─ __pycache__
 - └─ records
 - └─ templates
 - └─ __init__.py
 - └─ settings.py
 - └─ urls.py
 - └─ wsgi.py
 - └─ db.sqlite3
 - └─ manage.py
 - └─ include
 - └─ lib
 - └─ pip-selfcheck.json

GETTING STARTED WITH DJANGO

Define one or more view functions (one for each URL endpoint)

```
...  
  
def about(request):  
    return render(request,  
        'aboutme.html', {'name': 'John Doe',  
        'age': 20})  
  
def newContact(request):  
    ...  
  
def list(request):  
    ...  
  
def deleteContact(request):  
    ...
```



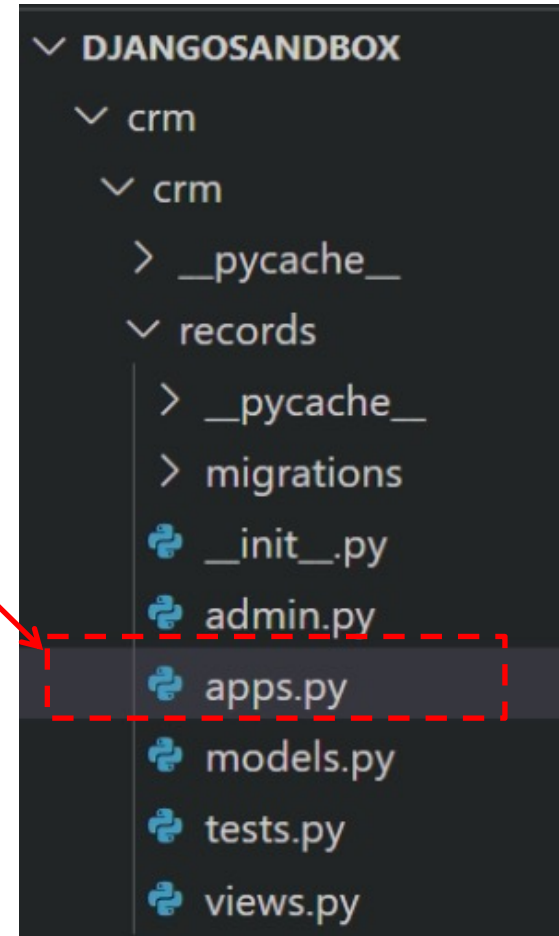
- └─ djangosandbox
 - └─ bin
 - └─ crm
 - └─ crm
 - └─ __pycache__
 - └─ records
 - └─ migrations
 - └─ __init__.py
 - └─ admin.py
 - └─ apps.py
 - └─ forms.py
 - └─ models.py
 - └─ tests.py
 - └─ **views.py**
 - └─ templates
 - └─ __init__.py
 - └─ settings.py
 - └─ urls.py
 - └─ wsgi.py
 - └─ db.sqlite3
 - └─ manage.py
 - └─ include
 - └─ lib
 - └─ pip-selfcheck.json

GETTING STARTED WITH DJANGO

```
from django.apps import AppConfig

class RecordsConfig(AppConfig):
    name = 'crm.records'
```

Since Django 3.2, we need to add full path (records is inside the **crm** module)



Just normal HTML
pages

DEFINING TEMPLATE

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>About me</title>
  </head>
  <body>
    <h1>Hi {{ name }}</h1>
    <h2>Age: {{ age }}</h2>
  </body>
</html>
```

▲ templates

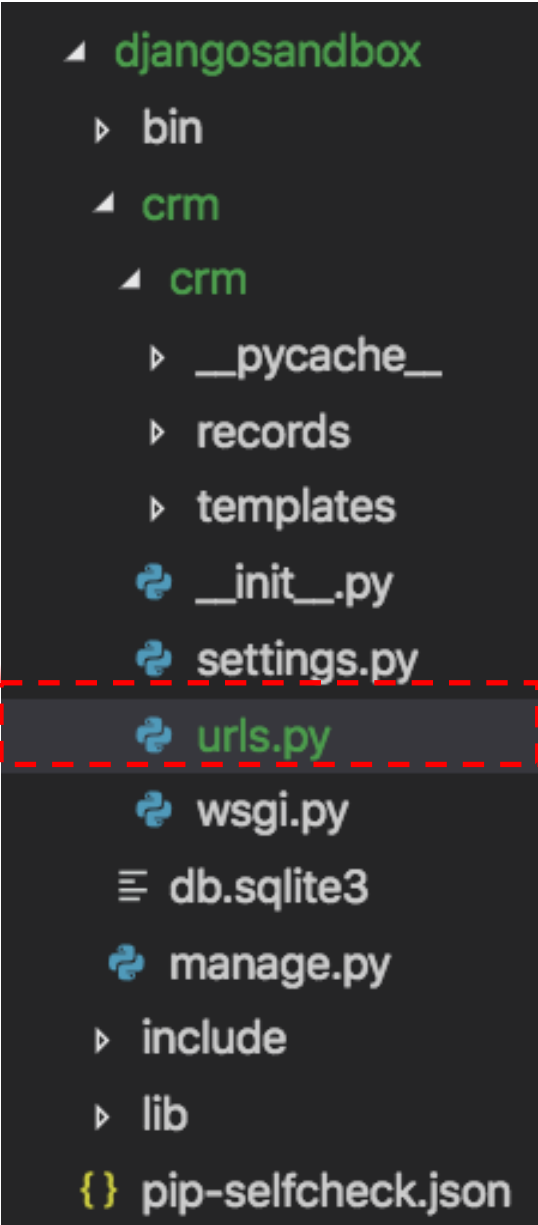
- <> aboutme.html
- <> homepage.html
- <> list.html
- <> new.html

Display data passed
from the view function

GETTING STARTED WITH DJANGO

```
...  
  
from crm.records import views as  
records_views  
  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('about/', records_views.about),  
    path('',  
        TemplateView.as_view(template_name='home.html'),  
    ],  
]
```

Add the URL endpoints and link with the view functions



- ▲ djangosandbox
 - bin
 - ▲ crm
 - ▲ crm
 - __pycache__
 - records
 - templates
 - 🔗 __init__.py
 - 🔗 settings.py
 - 🔗 urls.py**
 - 🔗 wsgi.py
 - ≡ db.sqlite3
 - 🔗 manage.py
 - include
 - lib
 - {} pip-selfcheck.json

GETTING STARTED WITH DJANGO

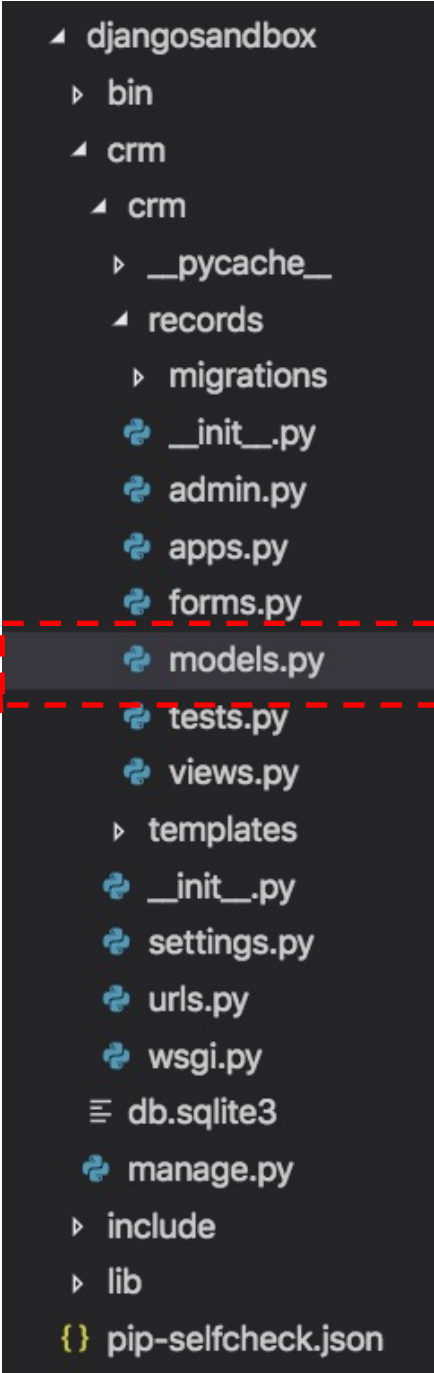
```
from django.db import models

# Create your models here.

class Customer(models.Model):
    name = models.CharField(max_length=30)
    age = models.IntegerField()

    def __str__(self):
        return f'name: {self.name}, age: {self.age}'
```

Define one or more models
to be used in views.py



- └─ djanglesandbox
 - └─ bin
 - └─ crm
 - └─ crm
 - └─ __pycache__
 - └─ records
 - └─ migrations
 - └─ __init__.py
 - └─ admin.py
 - └─ apps.py
 - └─ forms.py
 - └─ models.py
 - └─ tests.py
 - └─ views.py
 - └─ templates
 - └─ __init__.py
 - └─ settings.py
 - └─ urls.py
 - └─ wsgi.py
 - └─ db.sqlite3
 - └─ manage.py
 - └─ include
 - └─ lib
 - └─ pip-selfcheck.json

GETTING STARTED WITH DJANGO

Run Django migrate operation to create the customer table

- `python manage.py makemigrations`
- `python manage.py migrate`

```
(djangosandbox) NUSs-MacBook-Pro-93:crm lekhsian$ python manage.py makemigrations
Migrations for 'records':
  crm/records/migrations/0001_initial.py
    - Create model Customer
(djangosandbox) NUSs-MacBook-Pro-93:crm lekhsian$ python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, records, sessions
Running migrations:
  Applying records.0001_initial... OK
```

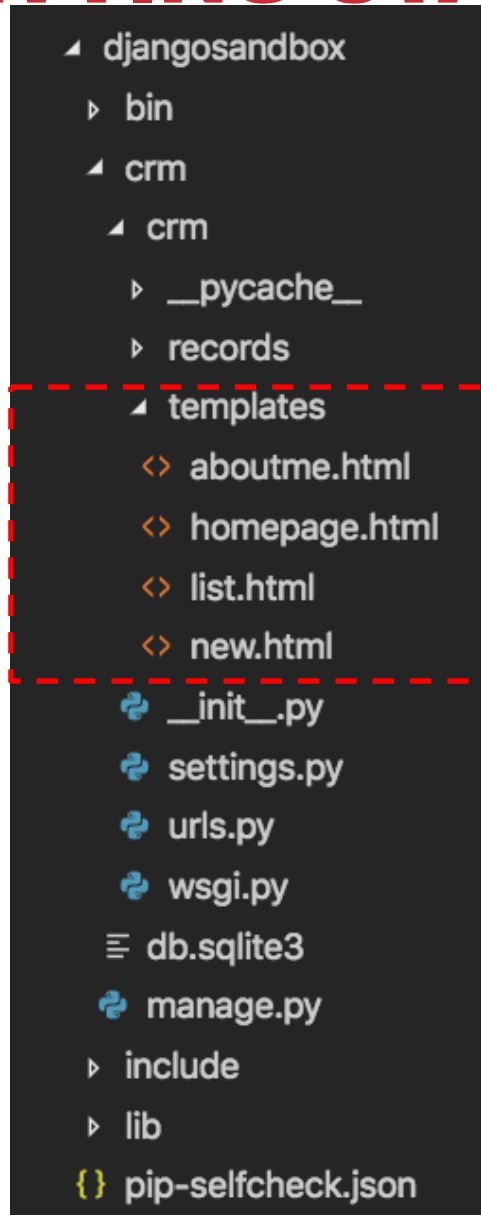
GETTING STARTED WITH DJANGO

```
...  
  
from crm.records import views as  
records_views  
  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('about/', records_views.about),  
    path('new/',  
        records_views.newContact),  
    path('list/', records_views.list),  
    path('delete/',  
        records_views.deleteContact),  
    path('',  
        TemplateView.as_view(template_name='home  
page.html')),  
]
```

Add the URL endpoints and
link with the view functions

- ▲ djangosandbox
 - bin
 - ▲ crm
 - ▲ crm
 - __pycache__
 - records
 - templates
 - 🔗 __init__.py
 - 🔗 settings.py
 - 🔗 **urls.py**
 - 🔗 wsgi.py
 - ≡ db.sqlite3
 - 🔗 manage.py
 - include
 - lib
 - {} pip-selfcheck.json

GETTING STARTED WITH DJANGO



Add any additional
templates

New Entry

127.0.0.1:8000/new/

Search

Most Visited NUS Library New Tab

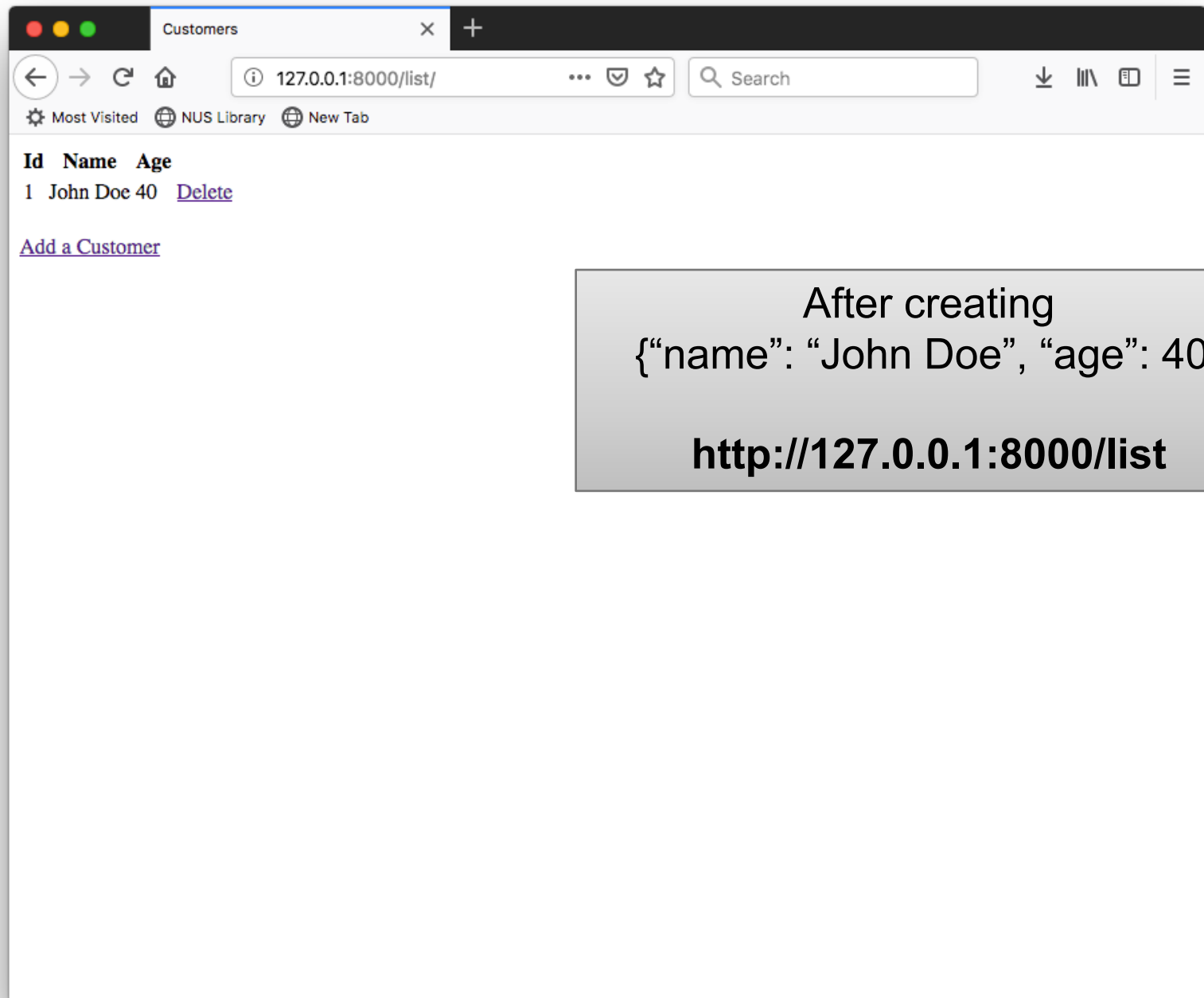
New Customer

Name:

Age:

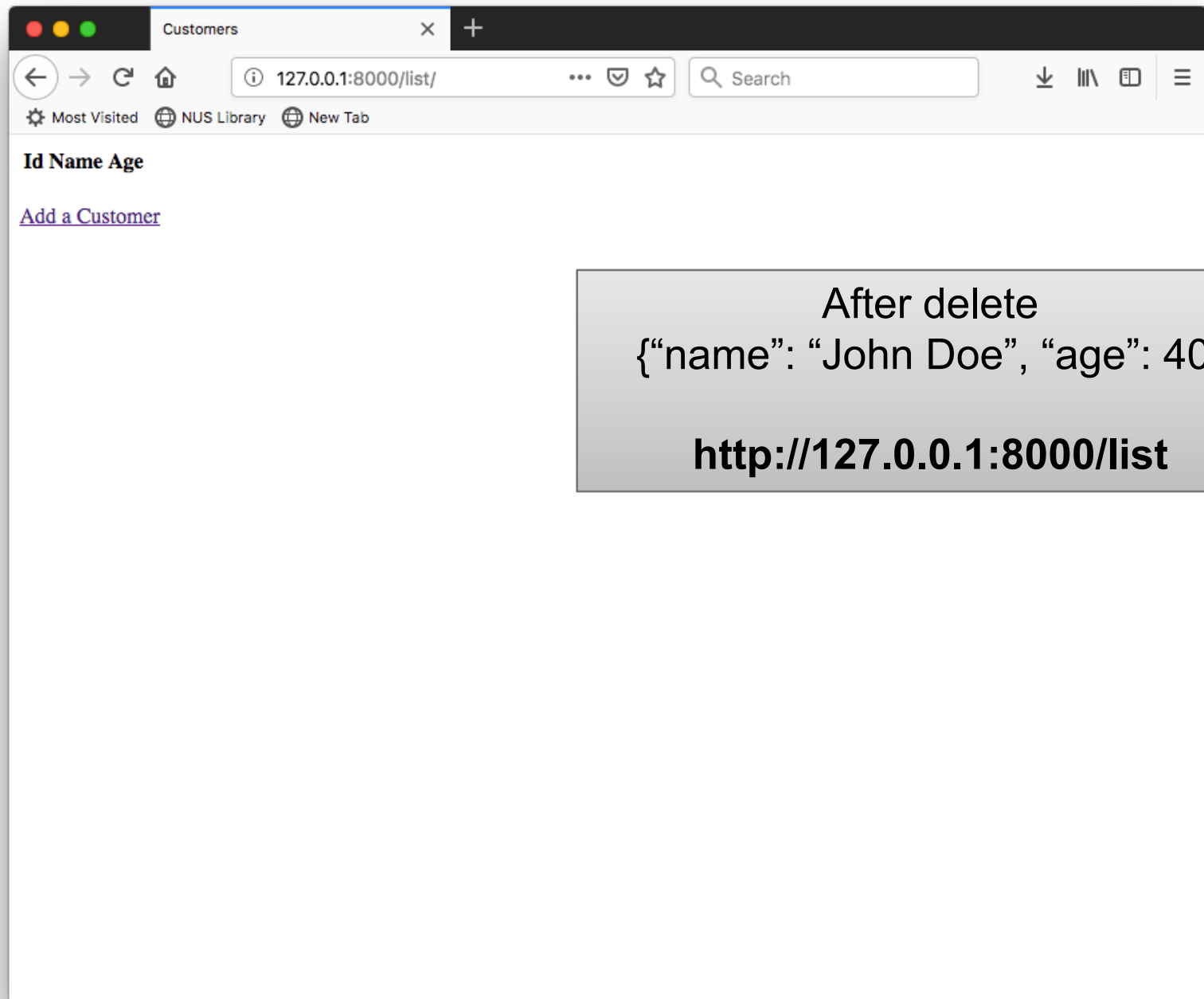
If everything works out
(skipped some files --- check
out the sample project zip file)

<http://127.0.0.1:8000/new>



After creating
{“name”: “John Doe”, “age”: 40}

<http://127.0.0.1:8000/list>



WHAT'S NEXT?

Enterprise System Architecture Design 2